

CASE STUDY

SMART TUNING FOR WASTEWATER

Multi-objective Bayesian optimization, trained on **1,352 days** of real treatment-plant data, mapped the trade-off between **energy use and effluent quality** in a **single optimization run**, giving operators a menu of settings instead of one guess.

Balancing Energy and Effluent Quality With Data

Wastewater treatment plants protect public health, but energy is one of their largest operating costs. Every setting is a trade-off: more aeration or retention time can lower pollutant levels, yet drives up energy use, and plant behaviour shifts with temperature and inflow. This project applied the same optimization engine Better Materials uses for material development to a process problem: tuning plant operations so energy and pollution fall together.

OVERVIEW

Effluent quality is judged by Biochemical and Chemical Oxygen Demand (BOD and COD). Lowering them while cutting energy is a three-way balancing act with nonlinear, interacting controls.

Machine learning models learned how operating conditions drive energy, BOD and COD. A multi-objective Bayesian optimizer then searched for settings that improve all three, returning a set of best trade-offs rather than a single answer.

THE PROJECT

The study used a public dataset of 1,352 daily records from a full-scale wastewater treatment plant in Melbourne, Australia: inflow, outflow, temperature, hydraulic retention time, tank volume, grid energy, BOD and COD.

XGBoost models predicted each target, and sequential Bayesian optimization on the SigOpt platform, guided by an Upper Confidence Bound acquisition function, searched the operating space.

At a Glance

- **Data:** 1,352 days of operating records from a full-scale plant.
- **Controls:** inflow, outflow, temperature, retention time, volume and day.
- **Goals:** minimise grid energy, BOD and COD at the same time.
- **Method:** XGBoost predictions feeding multi-objective Bayesian optimization.
- **Output:** a Pareto front of best-balanced operating settings.



Cut Energy

Find operating settings that need less grid power.



Protect Water Quality

Keep BOD and COD within treatment targets while saving energy.



Choose the Trade-Off

Give operators a range of options to match changing priorities.

The Challenge

Plant controls interact in nonlinear ways. A small rise in inflow without matching changes to retention time or aeration can push BOD up; over-aeration cleans the water but wastes energy.

Classic approaches struggle at this scale. A full factorial test of just six parameters at four levels needs 4,096 runs, and genetic algorithms can take many generations to converge and depend heavily on their own tuning.

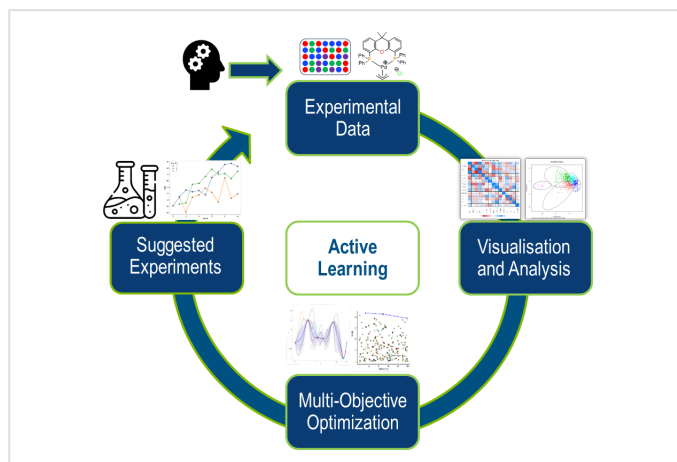
01	Clean and explore the data 1,352 daily records were checked, normalised and profiled to understand distributions and correlations.
02	Learn how the plant behaves Separate XGBoost models predicted energy, BOD and COD, trained on 80% of the data and tested on the rest.
03	Search with Bayesian optimization An Upper Confidence Bound strategy balanced exploring new settings with refining promising ones.
04	Deliver a menu of options The run returned Pareto-optimal settings, each a different balance of energy and effluent quality.

What It Means for Operators

- Energy and treatment quality are optimised together, not traded blindly.
- Operators pick settings by priority: save power or tighten quality.
- The same engine tunes material processing: mixing, curing and kiln conditions.

How It Works

An active-learning loop links data, analysis and optimization: each round of results sharpens the model, and the model suggests the next settings worth evaluating.



Results

Within a single optimization run, the framework produced a well-defined Pareto front. Settings with the lowest energy carried slightly higher BOD or COD, while the cleanest effluent needed moderately more energy, and several configurations kept both pollutants in range while using markedly less energy than baseline or random settings.

Energy model accuracy

R² 0.84

BOD 0.79 · COD 0.83

To a Pareto front

1 run

vs. 4,096 factorial tests

Model-based study. Suggested settings were evaluated with predictive models built from historical data. Applying them at a plant would require on-site validation first.

How Well the Models Learned

Predicted vs. actual values for BOD, energy and COD on training (blue) and test (red) data. The models explain most of the variation in all three targets, which is what makes their scores trustworthy guides for the optimizer.

0.841

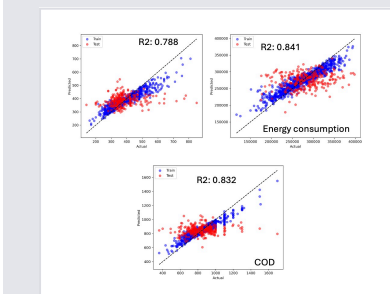
Energy R²

0.832

COD R²

0.788

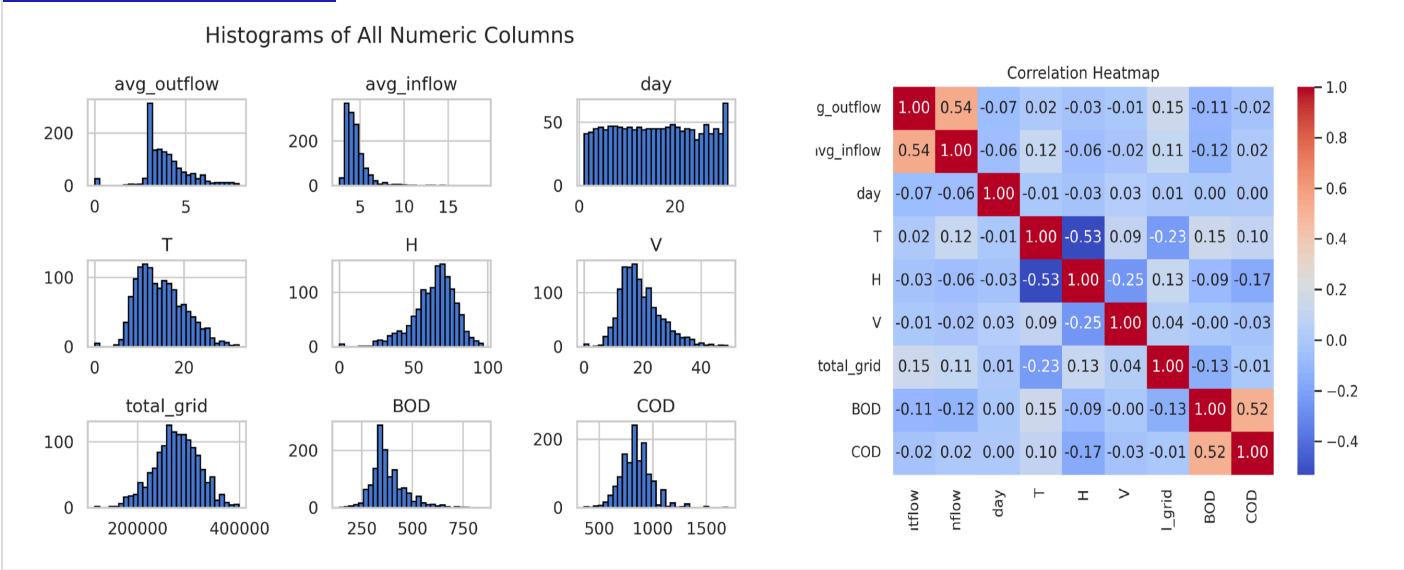
BOD R²



Results in Detail

From raw plant records to a ranked set of operating options: what the data looks like, how well the models learned it, and where the optimizer landed.

FIGURE 1 • THE PLANT DATA



Clean, varied, real-world data. Distributions of all nine variables (left) and their correlations (right). Temperature and retention time move in opposite directions, and BOD and COD track each other, the kinds of interactions a single-variable approach would miss.

FIGURE 2 • OPTIMIZATION RESULTS



A menu of best trade-offs. Each point is a suggested setting; the circled points form the Pareto front, with model-predicted grid energy of roughly 171,000 to 205,000, below the plant's historical average of about 277,000. Parameter importance (top right) shows outflow and tank volume mattered most for effluent quality, while day and inflow drove energy; the parallel-coordinates view (bottom) traces every suggestion across all inputs.